

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5101

5 Credits: 1.5

Program Core

Source-grounded

Packaging Techniques

□ To provide concrete knowledge on fundamentals of packaging technology.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5101 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5101.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Printing Technology
Course code	5101
Course title	Packaging Techniques
Semester	5 Credits: 1.5
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 45
Periods per semester	45
Credits	1.5
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

28 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES**Objectives, outcomes and mapping****Course objectives**

- To provide concrete knowledge on fundamentals of packaging technology.
- To guide the student to the basic concepts of the packaging like its functions, importance, need, medium, variations of packaging techniques, machines, legal-security and marketing aspects, packaging tests, trends etc.
- To think and understand the packaging concept together with the printing aspects.
- To develop idea on the packaging trends and exciting possibilities of packaging in

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding packaging. Explain about various packaging materials and	See official syllabus
CO2	14 Understanding related finishing operations. Extend knowledge on various packaging	See official syllabus
CO3	15 Understanding techniques. Extend knowledge on various packaging	See official syllabus
CO4	14 Understanding machineries and trends in packaging.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2

Row	Extracted official mapping
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Describe the importance and basic concepts of packaging.	7	As specified
Module 2	finishing operation.	6	As specified
Module 3	Extend knowledge on various packaging techniques.	9	As specified
Module 4	packaging.	6	As specified

MODULE 1

Describe the importance and basic concepts of packaging.

This module is presented from the official syllabus scope for Packaging Techniques. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe the importance and basic concepts of packaging.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

▼ Packaging: Introduction.

Packaging: Introduction. is an official topic in Module 1 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Packaging: Introduction. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, packaging: introduction. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Packaging: Introduction. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പാക്കേജിംഗ്: Introduction. പാക്കേജിംഗിന്റെ definition/meaning പാക്കേജിംഗിന്റെ. പാക്കേജിംഗ് പാക്കേജിംഗ് പാക്കേജിംഗ്, steps, application പാക്കേജിംഗ് പാക്കേജിംഗ് പാക്കേജിംഗ്. Technical terms English-ൽ പാക്കേജിംഗ് പാക്കേജിംഗ്.

▼ Functions of packaging

Functions of packaging is an official topic in Module 1 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Functions of packaging using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, functions of packaging should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Functions of packaging means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പാക്കേജിംഗിന്റെ പ്രവർത്തനങ്ങൾ
 പാക്കേജിംഗിന്റെ definition/meaning പാക്കേജിംഗിന്റെ പ്രവർത്തനങ്ങൾ. പാക്കേജിംഗിന്റെ പ്രവർത്തനങ്ങൾ, steps, application പാക്കേജിംഗിന്റെ പ്രവർത്തനങ്ങൾ. Technical terms English-ൽ പാക്കേജിംഗിന്റെ പ്രവർത്തനങ്ങൾ.

▼ Types of packaging

Types of packaging is an official topic in Module 1 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Types of packaging using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, types of packaging should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Types of packaging means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Types of packaging
 definition/meaning .
 application
 . Technical terms English-
 .

▼ Correlation of printing and packaging.

Correlation of printing and packaging. is an official topic in Module 1 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Correlation of printing and packaging. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, correlation of printing and packaging. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Correlation of printing and packaging. means in the official course context

▼ Package design concept.

Package design concept. is an official topic in Module 1 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Package design concept. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, package design concept. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Package design concept. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പാക്കേജ ഡിസൈൻ കോൺസെപ്റ്റ്. പാക്കേജ ഡിസൈൻ കോൺസെപ്റ്റ് definition/meaning പാക്കേജ ഡിസൈൻ കോൺസെപ്റ്റ്. പാക്കേജ ഡിസൈൻ കോൺസെപ്റ്റ്, steps, application പാക്കേജ ഡിസൈൻ കോൺസെപ്റ്റ് പാക്കേജ ഡിസൈൻ കോൺസെപ്റ്റ്. Technical terms English-പാക്കേജ ഡിസൈൻ കോൺസെപ്റ്റ്.

▼ Marking on packages.

Marking on packages. is an official topic in Module 1 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Marking on packages. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, marking on packages. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Marking on packages. means in the official course context
Study lens	Principle → components or stages → result → application

▼ **Tests on packages. 4 Understanding Packaging: Introduction- Definition, Importance of packaging. Functions of packaging - protection, utility, communication, containment. Types-Flexible packaging, Rigid packaging, Consumer packaging, Industrial packaging, Military packaging. Correlate the possibilities of printing with packaging. Factors influencing design of package. Markings on package - Handling marks, Routing marks, Information marks. Tests on package: Physical & Mechanical test- Drop test, Vibration test, Compression test, Inclined impact test, Rolling test, Drum test. Chemical & Climatic test - Rain test, Sand and Dust test, Salt Spray test, Fungus Resistance test, Cobb test. Explain about various packaging materials and packaging-oriented**

Tests on packages. 4 Understanding Packaging: Introduction- Definition, Importance of packaging. Functions of packaging - protection, utility, communication, containment. Types-Flexible packaging, Rigid packaging, Consumer packaging, Industrial packaging, Military packaging. Correlate the possibilities of printing with packaging. Factors influencing design of package. Markings on package - Handling marks, Routing marks, Information marks. Tests on package: Physical & Mechanical test- Drop test, Vibration test, Compression test, Inclined impact test, Rolling test, Drum test. Chemical & Climatic test - Rain test, Sand and Dust test, Salt Spray test, Fungus Resistance test, Cobb test. Explain about various packaging materials and packaging-oriented is an official topic in Module 1 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Tests on packages. 4 Understanding Packaging: Introduction- Definition, Importance of packaging. Functions of packaging - protection, utility, communication, containment. Types-Flexible packaging, Rigid packaging, Consumer packaging, Industrial packaging, Military packaging. Correlate the possibilities of printing with packaging. Factors influencing design of package. Markings on package - Handling marks, Routing marks, Information marks. Tests on package: Physical & Mechanical test- Drop test, Vibration test, Compression test, Inclined impact test, Rolling test, Drum test. Chemical & Climatic test - Rain test, Sand and Dust test, Salt Spray test, Fungus Resistance test, Cobb test. Explain about various packaging materials and packaging-oriented using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, tests on packages. 4 understanding packaging: introduction- definition, importance of packaging. functions of packaging - protection, utility, communication, containment. types-flexible packaging, rigid packaging, consumer packaging, industrial packaging, military packaging. correlate the possibilities of printing with packaging. factors influencing design of package. markings on package - handling marks, routing marks, information marks. tests on package: physical & mechanical test- drop test, vibration test, compression test, inclined impact test, rolling test, drum test. chemical & climatic test - rain test, sand and dust test, salt spray test, fungus resistance test, cobb test. explain about various packaging materials and packaging-oriented should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

definition/meaning പദപരിഭാഷണം. പദപരിഭാഷണം പദപരിഭാഷണം, steps, application പദപരിഭാഷണം പദപരിഭാഷണം പദപരിഭാഷണം. Technical terms English-പദപരിഭാഷണം പദപരിഭാഷണം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2**finishing operation.**

This module is presented from the official syllabus scope for Packaging Techniques. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: finishing operation.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ Packaging materials.

Packaging materials. is an official topic in Module 2 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Packaging materials. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, packaging materials. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Packaging materials. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പാക്കേജിംഗ് മെറ്റീരിയലുകൾ. പാക്കേജിംഗ് മെറ്റീരിയലുകളുടെ definition/meaning പാക്കേജിംഗ് മെറ്റീരിയലുകൾ. പാക്കേജിംഗ് മെറ്റീരിയലുകൾ, steps, application പാക്കേജിംഗ് മെറ്റീരിയലുകൾ പാക്കേജിംഗ്. Technical terms English-ൽ പാക്കേജിംഗ് മെറ്റീരിയലുകൾ.

▼ Cushioning Materials

Cushioning Materials is an official topic in Module 2 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Cushioning Materials using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, cushioning materials should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Cushioning Materials means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പാദങ്ങളിലെ പാദങ്ങളുടെ പദപരിഭാഷ, definition/meaning പദപരിഭാഷ. പദപരിഭാഷ പദപരിഭാഷ, steps, application പദപരിഭാഷ പദപരിഭാഷ. Technical terms English-ൽ പദപരിഭാഷ പദപരിഭാഷ.

▼ Adhesives.

Adhesives. is an official topic in Module 2 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Adhesives. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, adhesives. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Adhesives. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പ്രദർശനം Adhesives. പ്രദർശനം പ്രദർശനം definition/meaning പ്രദർശനം പ്രദർശനം. പ്രദർശനം പ്രദർശനം പ്രദർശനം, steps, application പ്രദർശനം പ്രദർശനം പ്രദർശനം. Technical terms English-പ്രദർശനം പ്രദർശനം പ്രദർശനം.

▼ Adhesive tapes.

Adhesive tapes. is an official topic in Module 2 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Adhesive tapes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, adhesive tapes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Adhesive tapes. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പ്രശ്നം Adhesive tapes. പദപരിഭാഷണം പദപരിഭാഷണം definition/meaning പദപരിഭാഷണം. പദപരിഭാഷണം പദപരിഭാഷണം, steps, application പദപരിഭാഷണം പദപരിഭാഷണം പദപരിഭാഷണം. Technical terms English-പദപരിഭാഷണം പദപരിഭാഷണം.

▼ Closures. 2 Understanding Finishing operations - Lamination, Coating,

Closures. 2 Understanding Finishing operations - Lamination, Coating, is an official topic in Module 2 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Closures. 2 Understanding Finishing operations - Lamination, Coating, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, closures. 2 understanding finishing operations - lamination, coating, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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▼ 4 Understanding Varnishing, Sealing and Labeling. Packaging materials: Wood, Board, Glass, Plastic, Metals, Glass, Corrugated board-Paper, Plastic and Fiber Board. Cushioning Materials - Functions, Properties. Adhesive: Functions, Properties, Types - Natural and Synthetic, Solvent Responsive Adhesive, Pressure Sensitive Adhesive, Heat sensitive adhesive, Chemically reactive adhesive. Adhesive tapes- fabric tapes, Paper tapes, Film tapes, Foil tapes, Foam tapes, Two faced tapes. Closure: Features and functions and types. Finishing operations: Definition, introduction, Lamination: Functions, Properties, Types - Wet lamination, Dry lamination, Hot melts lamination, Extrusion lamination. Coatings - Functions, Properties, Types - Primer coating, Protective coating, Heat sealable and Self adhesive coating. Varnishing: Features, importance and Types - Oil Varnish, Synthetic Varnishing, Spirit varnish, Cellulose varnish, Water varnish, Roller coating, Lacquers. Sealing: Definition, purpose, Methods- Heat sealing, Cold sealing. Labeling: definition, purpose, basic elements of labeling.

4 Understanding Varnishing, Sealing and Labeling. Packaging materials: Wood, Board, Glass, Plastic, Metals, Glass, Corrugated board-Paper, Plastic and Fiber Board. Cushioning Materials - Functions, Properties. Adhesive: Functions, Properties, Types - Natural and Synthetic, Solvent Responsive Adhesive, Pressure Sensitive Adhesive, Heat sensitive adhesive, Chemically reactive adhesive. Adhesive tapes-fabric tapes, Paper tapes, Film tapes, Foil tapes, Foam tapes, Two faced tapes. Closure: Features and functions and types. Finishing operations: Definition, introduction, Lamination: Functions, Properties, Types - Wet lamination, Dry lamination, Hot melts lamination, Extrusion lamination. Coatings - Functions, Properties, Types - Primer coating, Protective coating, Heat sealable and Self adhesive coating. Varnishing: Features, importance and Types - Oil Varnish, Synthetic Varnishing, Spirit varnish, Cellulose varnish, Water varnish, Roller coating, Lacquers. Sealing: Definition, purpose, Methods- Heat sealing, Cold sealing. Labeling: definition, purpose, basic elements of labeling. is an official topic in Module 2 of Packaging Techniques. Study the

terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding Varnishing, Sealing and Labeling. Packaging materials: Wood, Board, Glass, Plastic, Metals, Glass, Corrugated board-Paper, Plastic and Fiber Board. Cushioning Materials - Functions, Properties. Adhesive: Functions, Properties, Types - Natural and Synthetic, Solvent Responsive Adhesive, Pressure Sensitive Adhesive, Heat sensitive adhesive, Chemically reactive adhesive. Adhesive tapes-fabric tapes, Paper tapes, Film tapes, Foil tapes, Foam tapes, Two faced tapes. Closure: Features and functions and types. Finishing operations: Definition, introduction, Lamination: Functions, Properties, Types - Wet lamination, Dry lamination, Hot melts lamination, Extrusion lamination. Coatings - Functions, Properties, Types - Primer coating, Protective coating, Heat sealable and Self adhesive coating. Varnishing: Features, importance and Types - Oil Varnish, Synthetic Varnishing, Spirit varnish, Cellulose varnish, Water varnish, Roller coating, Lacquers. Sealing: Definition, purpose, Methods- Heat sealing, Cold sealing. Labeling: definition, purpose, basic elements of labeling. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding varnishing, sealing and labeling. packaging materials: wood, board, glass, plastic, metals, glass, corrugated board-paper, plastic and fiber board. cushioning materials - functions, properties. adhesive: functions, properties, types - natural and synthetic, solvent responsive adhesive, pressure sensitive adhesive, heat sensitive adhesive, chemically reactive adhesive. adhesive tapes-fabric tapes, paper tapes, film tapes, foil tapes, foam tapes, two faced tapes. closure: features and functions and types. finishing operations: definition, introduction, lamination: functions, properties, types - wet lamination, dry lamination, hot melts lamination, extrusion lamination. coatings - functions, properties, types - primer coating, protective coating, heat sealable and self adhesive coating. varnishing: features, importance and types - oil varnish, synthetic varnishing, spirit varnish, cellulose varnish, water varnish, roller coating, lacquers. sealing: definition, purpose, methods- heat sealing, cold sealing. labeling: definition, purpose, basic elements of labeling. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding Varnishing, Sealing and Labeling. Packaging materials: Wood, Board, Glass, Plastic, Metals, Glass, Corrugated board-Paper, Plastic and Fiber Board. Cushioning Materials - Functions, Properties. Adhesive: Functions, Properties, Types - Natural and Synthetic, Solvent Responsive Adhesive, Pressure Sensitive Adhesive, Heat sensitive adhesive, Chemically reactive adhesive. Adhesive tapes-fabric tapes, Paper tapes, Film tapes, Foil tapes, Foam tapes, Two faced tapes. Closure: Features and functions and types. Finishing operations: Definition, introduction, Lamination: Functions, Properties, Types - Wet lamination, Dry lamination, Hot melts lamination, Extrusion lamination. Coatings - Functions, Properties, Types - Primer coating, Protective coating, Heat sealable and Self adhesive coating. Varnishing: Features, importance and Types - Oil Varnish, Synthetic Varnishing, Spirit varnish, Cellulose varnish, Water varnish, Roller coating, Lacquers. Sealing: Definition, purpose, Methods- Heat sealing, Cold sealing. Labeling: definition, purpose, basic elements of labeling. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-4 Understanding Varnishing, Sealing and Labeling. Packaging materials: Wood, Board, Glass, Plastic, Metals, Glass, Corrugated board-Paper, Plastic and Fiber Board. Cushioning Materials - Functions, Properties. Adhesive: Functions, Properties, Types - Natural and Synthetic, Solvent Responsive Adhesive, Pressure Sensitive Adhesive, Heat sensitive adhesive, Chemically reactive adhesive. Adhesive tapes-fabric tapes, Paper tapes, Film tapes, Foil tapes, Foam tapes, Two faced tapes. Closure: Features and functions and types. Finishing operations: Definition, introduction, Lamination: Functions, Properties, Types - Wet lamination, Dry lamination, Hot melts lamination, Extrusion lamination. Coatings - Functions, Properties, Types - Primer coating, Protective coating, Heat sealable and Self adhesive coating. Varnishing: Features, importance and Types - Oil Varnish, Synthetic Varnishing, Spirit varnish, Cellulose varnish, Water varnish, Roller coating, Lacquers. Sealing: Definition, purpose, Methods- Heat sealing, Cold sealing. Labeling: definition, purpose, basic elements of labeling.

ৱিকিপিডিয়াৰ পৰা definition/meaning ৰ পৰা লৈয়াহৈছে। ৱিকিপিডিয়াৰ পৰা লৈয়াহৈছে, steps, application ৰ পৰা লৈয়াহৈছে ৱিকিপিডিয়াৰ পৰা লৈয়াহৈছে। Technical terms English-ৰ পৰা লৈয়াহৈছে ৱিকিপিডিয়াৰ পৰা লৈয়াহৈছে।

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Extend knowledge on various packaging techniques.

This module is presented from the official syllabus scope for Packaging Techniques. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Extend knowledge on various packaging techniques.
- Official duration: As specified
- Official topic entries captured: 9
- The full source excerpt is preserved below for coverage checking.

▼ Thermoforming.

Thermoforming. is an official topic in Module 3 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Thermoforming. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, thermoforming. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Thermoforming. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എന്ന തെർമോഫോമിംഗ്. എന്താണ് തെർമോഫോമിംഗ്? definition/meaning എന്താണ് തെർമോഫോമിംഗ്. എന്താണ് തെർമോഫോമിംഗ്, steps, application എന്താണ് തെർമോഫോമിംഗ്. Technical terms English-എന്ന തെർമോഫോമിംഗ്.

▼ Molding.

Molding. is an official topic in Module 3 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Molding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, molding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Molding. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പ്ലാസ്റ്റിക് മോൾഡിംഗ്. പ്ലാസ്റ്റിക് മോൾഡിംഗിന്റെ നിർവ്വചനം/അർത്ഥം പ്ലാസ്റ്റിക് മോൾഡിംഗിന്റെ പ്രയോഗം പ്ലാസ്റ്റിക് മോൾഡിംഗിന്റെ ഘട്ടങ്ങൾ, steps, application പ്ലാസ്റ്റിക് മോൾഡിംഗിന്റെ ഉദാഹരണം. Technical terms English-ൽ പ്ലാസ്റ്റിക് മോൾഡിംഗിന്റെ നിർവ്വചനം.

▼ Thermoform packaging.

Thermoform packaging. is an official topic in Module 3 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Thermoform packaging. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, thermoform packaging. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Thermoform packaging. means in the official course context
Study lens	Principle → components or stages → result → application

▼ Stretch Wrapping & Strip packaging.

Stretch Wrapping & Strip packaging. is an official topic in Module 3 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Stretch Wrapping & Strip packaging. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, stretch wrapping & strip packaging. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Stretch Wrapping & Strip packaging. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-സ്റ്റച്ച് വ്രാപ്പിംഗ് & റിപ്പ് പാക്കേജിംഗ്. സ്റ്റച്ച് വ്രാപ്പിംഗിന്റെ നിർവ്വചനം/അർത്ഥം എന്താണ്. സ്റ്റച്ച് വ്രാപ്പിംഗിന്റെ ഘട്ടങ്ങൾ, steps, application എന്താണ് സ്റ്റച്ച് വ്രാപ്പിംഗിന്റെ ഉപയോഗം. Technical terms English-ൽ എങ്ങനെ എഴുതണം.

▼ Active and Passive packaging

Active and Passive packaging is an official topic in Module 3 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Active and Passive packaging using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, active and passive packaging should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Active and Passive packaging means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-☐☐ Active and Passive packaging

000000000000 00000 definition/meaning 000000000000. 000000 000000
 00000000, steps, application 000000 0000000000 000000. Technical terms
 English-0 00000 000000000000.

▼ Aerosol packaging.

Aerosol packaging. is an official topic in Module 3 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Aerosol packaging. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, aerosol packaging. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Aerosol packaging. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓസറോൾ പാക്കേജിംഗ്. ഓസറോൾ പാക്കേജിംഗ് definition/meaning ഓസറോൾ പാക്കേജിംഗ്. ഓസറോൾ പാക്കേജിംഗ് steps, application ഓസറോൾ പാക്കേജിംഗ് ഓസറോൾ. Technical terms English-ഓസറോൾ പാക്കേജിംഗ്.

▼ Aseptic packaging.

Aseptic packaging. is an official topic in Module 3 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Aseptic packaging. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, aseptic packaging. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Aseptic packaging. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എസെപ്റ്റിക് പാക്കേജിംഗ്. എസെപ്റ്റിക് പാക്കേജിംഗ് definition/meaning എസെപ്റ്റിക് പാക്കേജിംഗ്. എസെപ്റ്റിക് പാക്കേജിംഗ് steps, application എസെപ്റ്റിക് പാക്കേജിംഗ്. Technical terms English-എസെപ്റ്റിക് പാക്കേജിംഗ്.

▼ Intelligent packaging.

Intelligent packaging. is an official topic in Module 3 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Intelligent packaging. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, intelligent packaging. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Intelligent packaging. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഇന്റലിജന്റ് പാക്കേജിംഗ്. ഇതിന്റെ definition/meaning വിശദീകരിക്കുക. ഇതിന്റെ പ്രയോഗം, steps, application വിശദീകരിക്കുക. Technical terms English-ൽ വിശദീകരിക്കുക.

▼ **Advanced packaging techniques. 1 Understanding Thermoforming: Vacuum forming, Drape forming, Snap back forming, Plug assist forming, Pressure forming. Molding- Compression Molding, Transfer Molding, Injection blow molding, Extrusion blow molding. Thermoform packaging: Skin packaging, Shrink packaging, and Blister packaging. Stretch Wrapping, Strip packaging. Active packaging: Vacuum packaging, Gas packaging. Passive Atmosphere Modification- MAP, CAP. Aerosol packaging- Features, importance, Components and Filling methods. Aseptic packaging- Features, importance, Components and Filling methods. Intelligent Packaging. Advanced packaging techniques-overview. Extend knowledge on various packaging machineries and trends in**

Advanced packaging techniques. 1 Understanding Thermoforming: Vacuum forming, Drape forming, Snap back forming, Plug assist forming, Pressure forming. Molding- Compression Molding, Transfer Molding, Injection blow molding, Extrusion blow molding. Thermoform packaging: Skin packaging, Shrink packaging, and Blister packaging. Stretch Wrapping, Strip packaging. Active packaging: Vacuum packaging, Gas packaging. Passive Atmosphere Modification- MAP, CAP. Aerosol packaging- Features, importance, Components and Filling methods. Aseptic packaging- Features, importance, Components and Filling methods. Intelligent Packaging. Advanced packaging techniques- overview. Extend knowledge on various packaging machineries and trends in is an official topic in Module 3 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Advanced packaging techniques. 1 Understanding Thermoforming: Vacuum forming, Drape forming, Snap back forming, Plug assist forming, Pressure forming. Molding- Compression Molding, Transfer Molding, Injection blow molding, Extrusion blow molding. Thermoform packaging: Skin packaging, Shrink packaging, and Blister packaging. Stretch Wrapping, Strip packaging. Active packaging: Vacuum packaging, Gas packaging. Passive Atmosphere Modification- MAP, CAP. Aerosol packaging- Features, importance, Components and Filling methods. Aseptic packaging- Features, importance, Components and Filling methods. Intelligent Packaging. Advanced packaging techniques- overview. Extend knowledge on various packaging machineries and trends in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, advanced packaging techniques. 1 understanding thermoforming: vacuum forming, drape forming, snap back forming, plug assist forming, pressure forming. molding- compression molding, transfer molding, injection blow molding, extrusion blow molding. thermoform packaging: skin packaging, shrink packaging, and blister packaging. stretch wrapping, strip packaging. active packaging: vacuum packaging, gas packaging. passive atmosphere modification- map, cap. aerosol packaging- features, importance, components and filling methods. aseptic packaging- features, importance, components and filling methods. intelligent packaging. advanced packaging techniques- overview. extend knowledge on various packaging machineries and trends in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Advanced packaging techniques. 1 Understanding Thermoforming: Vacuum forming, Drape forming, Snap back forming, Plug assist forming, Pressure forming. Molding- Compression Molding, Transfer Molding, Injection blow molding, Extrusion blow molding. Thermoform packaging: Skin packaging, Shrink packaging, and Blister packaging. Stretch Wrapping, Strip packaging. Active packaging: Vacuum packaging, Gas packaging. Passive Atmosphere Modification- MAP, CAP. Aerosol packaging- Features, importance, Components and Filling methods. Aseptic packaging- Features, importance, Components and Filling methods. Intelligent Packaging. Advanced packaging techniques- overview. Extend knowledge on various packaging machineries and trends in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Advanced packaging techniques. 1 Understanding Thermoforming: Vacuum forming, Drape forming, Snap back forming, Plug assist forming, Pressure forming. Molding- Compression Molding, Transfer Molding, Injection blow molding, Extrusion blow molding. Thermoform packaging: Skin packaging, Shrink packaging, and Blister packaging. Stretch Wrapping, Strip packaging. Active packaging: Vacuum packaging, Gas packaging. Passive Atmosphere Modification- MAP, CAP. Aerosol packaging- Features, importance, Components and Filling methods. Aseptic packaging- Features, importance, Components and Filling methods. Intelligent Packaging. Advanced packaging techniques- overview. Extend knowledge on various packaging machineries and trends in □□□□□□□□□□

രണ്ടാം definition/meaning രണ്ടാം രണ്ടാം രണ്ടാം, steps, application രണ്ടാം രണ്ടാം രണ്ടാം. Technical terms English- രണ്ടാം രണ്ടാം രണ്ടാം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4**packaging.**

This module is presented from the official syllabus scope for Packaging Techniques. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: packaging.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ Diverse aspects of various Packaging machines.

Diverse aspects of various Packaging machines. is an official topic in Module 4 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Diverse aspects of various Packaging machines. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, diverse aspects of various packaging machines. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Diverse aspects of various Packaging machines. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-☐☐ Diverse aspects of various Packaging machines. ☐☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐☐☐☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐☐☐☐☐☐☐☐☐☐☐☐☐☐.

▼ Carton making.

Carton making. is an official topic in Module 4 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Carton making. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, carton making. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Carton making. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

▼ Corrugated board Manufacturing.

Corrugated board Manufacturing. is an official topic in Module 4 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Corrugated board Manufacturing. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, corrugated board manufacturing. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Corrugated board Manufacturing. means in the official course context

▼ Box manufacturing.

Box manufacturing. is an official topic in Module 4 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Box manufacturing. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, box manufacturing. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Box manufacturing. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

▼ Packaging rules and regulations.

Packaging rules and regulations. is an official topic in Module 4 of Packaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Packaging rules and regulations. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, packaging rules and regulations. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Packaging rules and regulations. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പാക്കേജിംഗ് നിയമങ്ങൾ.

പാക്കേജിംഗ് നിയമങ്ങൾ definition/meaning പാക്കേജിംഗ് നിയമങ്ങൾ. പാക്കേജിംഗ് നിയമങ്ങൾ, steps, application പാക്കേജിംഗ് നിയമങ്ങൾ പാക്കേജിംഗ് നിയമങ്ങൾ. Technical terms English-ൽ പാക്കേജിംഗ് നിയമങ്ങൾ പാക്കേജിംഗ് നിയമങ്ങൾ.

▼ **Innovative trends in packaging. 1 Understanding Component description, purpose and basic working principles of various Packaging machines. Carton making-Introduction, design terminology, carton making procedures. Corrugated board Manufacturing: Materials, components and process. Box manufacturing- box styles, box designs, joints, closing styles. Packaging rules and regulations. FDA, FPO, MFPO. Innovative trends in packaging. Text / Reference: T/R Book Title/Author T1 IIP- Packaging technology- Vol - I, II, III- IIP India . R1 Frank Paine- Packaging design and performance. R2 Jhonn Briston- Advance in plastic packaging technology. Kit L. Yam-Encyclopedia of Packaging Technology- A John Wiley & Sons, R3 Inc., Publication. Online resources: Sl.No Website Link 1 <http://printwiki.org/Packing> 2 https://www.google.co.in/books/edition/Packaging_Technology 3 <https://www.packagingtechtoday.com/> 4 https://iip-in.com/about_iip/about-iip.aspx 5 <https://foodpackaging.foodtechconferences.org/>**

Innovative trends in packaging. 1 Understanding Component description, purpose and basic working principles of various Packaging machines. Carton making-Introduction, design terminology, carton making procedures. Corrugated board Manufacturing: Materials, components and process. Box manufacturing- box styles, box designs, joints, closing styles. Packaging rules and regulations. FDA, FPO, MFPO. Innovative trends in packaging. Text / Reference: T/R Book Title/Author T1 IIP- Packaging technology- Vol - I, II, III- IIP India . R1 Frank Paine- Packaging design and performance. R2 Jhonn Briston- Advance in plastic packaging technology. Kit L. Yam-Encyclopedia of Packaging Technology- A John Wiley & Sons, R3 Inc., Publication. Online resources: Sl.No Website Link 1 <http://printwiki.org/Packing> 2 https://www.google.co.in/books/edition/Packaging_Technology 3 <https://www.packagingtechtoday.com/> 4 https://iip-in.com/about_iip/about-iip.aspx 5 <https://foodpackaging.foodtechconferences.org/> is an official topic in Module 4 of Packaging Techniques. Study the terminology first,

then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Innovative trends in packaging. 1 Understanding Component description, purpose and basic working principles of various Packaging machines. Carton making-Introduction, design terminology, carton making procedures. Corrugated board Manufacturing: Materials, components and process. Box manufacturing- box styles, box designs, joints, closing styles. Packaging rules and regulations. FDA, FPO, MFPO. Innovative trends in packaging. Text / Reference: T/R Book Title/Author T1 IIP- Packaging technology- Vol – I, II, III- IIP India . R1 Frank Paine- Packaging design and performance. R2 Jhonn Briston- Advance in plastic packaging technology. Kit L. Yam-Encyclopedia of Packaging Technology- A John Wiley & Sons, R3 Inc., Publication. Online resources: Sl.No Website Link 1 <http://printwiki.org/Packing> 2 https://www.google.co.in/books/edition/Packaging_Technology 3 <https://www.packagingtechtoday.com/> 4 https://iip-in.com/about_iip/about-iip.aspx 5 <https://foodpackaging.foodtechconferences.org/> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, innovative trends in packaging. 1 understanding component description, purpose and basic working principles of various packaging machines. carton making-introduction, design terminology, carton making procedures. corrugated board manufacturing: materials, components and process. box manufacturing- box styles, box designs, joints, closing styles. packaging rules and regulations. fda, fpo, mfpo. innovative trends in packaging. text / reference: t/r book title/author t1 iip- packaging technology- vol – i, ii, iii- iip india . r1 frank paine- packaging design and performance. r2 jhonn briston- advance in plastic packaging technology. kit l. yam-encyclopedia of packaging technology- a john wiley & sons, r3 inc., publication. online resources: sl.no website link 1 <http://printwiki.org/packing> 2 https://www.google.co.in/books/edition/packaging_technology 3 <https://www.packagingtechtoday.com/> 4 https://iip-in.com/about_iip/about-iip.aspx 5 <https://foodpackaging.foodtechconferences.org/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Innovative trends in packaging. 1 Understanding Component description, purpose and basic working principles of various Packaging machines. Carton making-Introduction, design terminology, carton making procedures. Corrugated board Manufacturing: Materials, components and process. Box manufacturing- box styles, box designs, joints, closing styles. Packaging rules and regulations. FDA, FPO, MFPO. Innovative trends in packaging. Text / Reference: T/R Book Title/Author T1 IIP- Packaging technology- Vol – I, II, III- IIP India . R1 Frank Paine- Packaging design and performance. R2 Jhonn Briston- Advance in plastic packaging technology. Kit L. Yam-Encyclopedia of Packaging Technology- A John Wiley & Sons, R3 Inc., Publication. Online resources: Sl.No Website Link 1 http://printwiki.org/Packing 2 https://www.google.co.in/books/edition/Packaging_Technology 3 https://www.packagingtechtoday.com/ 4 https://iip-in.com/about_iip/about-iip.aspx 5 https://foodpackaging.foodtechconferences.org/ means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഇന്നിഷ്യേറ്റീവ് ട്രെൻഡ്സ് ഇൻ പാക്കേജിംഗ്. 1 Understanding Component description, purpose and basic working principles of various Packaging machines. Carton making-Introduction, design terminology, carton making procedures. Corrugated board Manufacturing: Materials, components and process. Box manufacturing- box styles, box designs, joints, closing styles. Packaging rules and regulations. FDA, FPO, MFPO. Innovative trends in packaging. Text / Reference: T/R Book Title/Author T1 IIP- Packaging technology- Vol – I, II, III- IIP India . R1 Frank Paine- Packaging design and performance. R2 Jhonn Briston- Advance in plastic packaging technology. Kit L. Yam-Encyclopedia of Packaging Technology- A John Wiley & Sons, R3 Inc., Publication. Online resources: Sl.No Website Link 1 <http://printwiki.org/Packing> 2 https://www.google.co.in/books/edition/Packaging_Technology 3 <https://www.packagingtechtoday.com/> 4 https://iip-in.com/about_iip/about-iip.aspx 5 <https://foodpackaging.foodtechconferences.org/> പാക്കേജിംഗ് എന്നതിന്റെ definition/meaning പാക്കേജിംഗ് എന്നതിന്റെ steps, application പാക്കേജിംഗ് എന്നതിന്റെ Technical terms English-ൽ പാക്കേജിംഗ് എന്നതിന്റെ.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

- ▶ **Define or explain Packaging: Introduction.. (3 marks)**
- ▶ **Define or explain Functions of packaging. (3 marks)**
- ▶ **Define or explain Types of packaging. (3 marks)**
- ▶ **Define or explain Correlation of printing and packaging.. (3 marks)**

Module 2

- ▶ **Define or explain Packaging materials.. (3 marks)**
- ▶ **Define or explain Cushioning Materials. (3 marks)**
- ▶ **Define or explain Adhesives.. (3 marks)**
- ▶ **Define or explain Adhesive tapes.. (3 marks)**

Module 3

- ▶ **Define or explain Thermoforming.. (3 marks)**
- ▶ **Define or explain Molding.. (3 marks)**

► **Define or explain Thermoform packaging.. (3 marks)**

► **Define or explain Stretch Wrapping & Strip packaging.. (3 marks)**

Module 4

► **Define or explain Diverse aspects of various Packaging machines.. (3 marks)**

► **Define or explain Carton making.. (3 marks)**

► **Define or explain Corrugated board Manufacturing.. (3 marks)**

► **Define or explain Box manufacturing.. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Packaging: Introduction..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Packaging materials..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Thermoforming..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram,

Module 4

1-mark definition: State the meaning of **Diverse aspects of various Packaging machines..**

3-mark explanation: Explain one principle, process or comparison from this module.

table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <http://printwiki.org/Packing>
- https://www.google.co.in/books/edition/Packaging_Technology
<https://www.packagingtechtoday.com/> https://iip-in.com/about_iip/about-iip.aspx
- <https://foodpackaging.foodtechconferences.org/>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5101. Verify institutional updates against the current official curriculum.